RA-PhyKSC-Diff: one reverse cold-diffusion iteration i

$x^{(i)}$



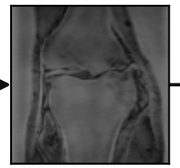
state at level t_i

1. Network prediction

(PhyKSC-Diff, frozen)

$$\hat{x}_0^{(i)} = f_\theta(x^{(i)}, t_i)$$

no retraining



$\hat{x}_0^{(i)}$

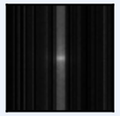
2. Fourier transform

$$\hat{k}_i = \mathcal{F}\hat{x}_0^{(i)}$$



3. Data consistency (first DC)

$$\hat{k}_i^{\mathrm{DC}} = M \odot y + (1 - M) \odot \hat{k}_i$$



y (acquired)



M (R4 mask)

4. Inverse FT

$$x_i^{\mathrm{DC}} = \mathcal{F}^{-1}\hat{k}_i^{\mathrm{DC}}$$



x_i^{DC}

5. Residual and step-size selection

reconstruction-change residual

$$R_i = \frac{\|x_i^{\mathrm{DC}} - x^{(i)}\|_2}{\|x^{(i)}\|_2 + \varepsilon}$$

relative change

$$\Delta R_i = \frac{|R_i - R_{i-1}|}{R_{i-1} + \varepsilon}$$

step size (levels)

$$\Delta t_i = 1 \quad \text{if } \Delta R_i > 0.02$$
$$\Delta t_i = 5 \quad \text{if } 0.005 < \Delta R_i \le 0.02$$
$$\Delta t_i = 10 \quad \text{if } \Delta R_i \le 0.005$$

$$t_{i+1} = \max(0,\ t_i - \Delta t_i)$$

example shown (knee test slice):

$t_i = 500,\ R_i = 0.087,\ \Delta R_i = 0.0049$

$\Rightarrow\ \Delta t_i = 10,\ t_{i+1} = 490$

6. Cold-diffusion step (jump)

$$\tilde{x}^{(i+1)} = x^{(i)} - \mathcal{F}^{-1} D_{t_i}\left(\hat{k}_i^{\mathrm{DC}}\right) + \mathcal{F}^{-1} D_{t_{i+1}}\left(\hat{k}_i^{\mathrm{DC}}\right)$$

re-introduces the columns removed between levels t_i and t_{i+1}; no extra network evaluation

7. Data consistency (second DC)

$$k^{(i+1)} = M \odot y + (1 - M) \odot \mathcal{F}\tilde{x}^{(i+1)}$$

acquired columns are unchanged by step 6; kept as a numerical safeguard

8. Inverse FT

$$x^{(i+1)} = \mathcal{F}^{-1} k^{(i+1)}$$

$x^{(i+1)}$



next state, level t_{i+1}

$i \leftarrow i + 1$

until $t_i = 0$

Blue: PhyKSC-Diff network with data consistency (as trained). NFE = number of iterations; the residual and the jump reuse tensors computed in steps 1-4.